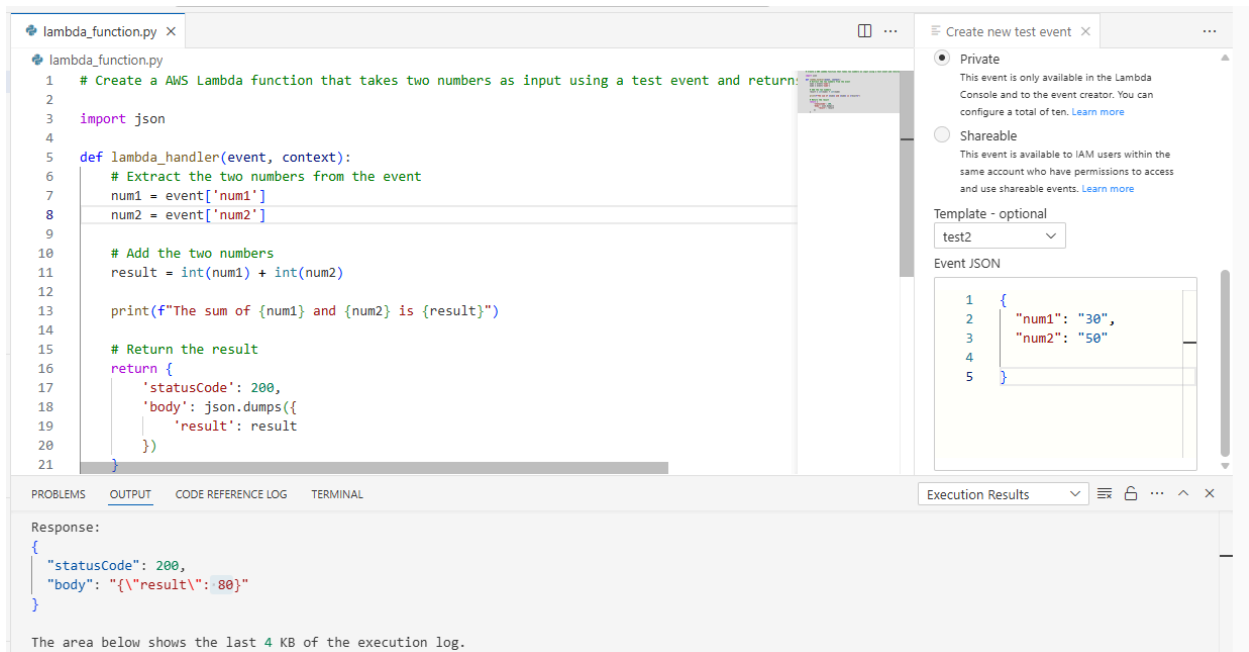
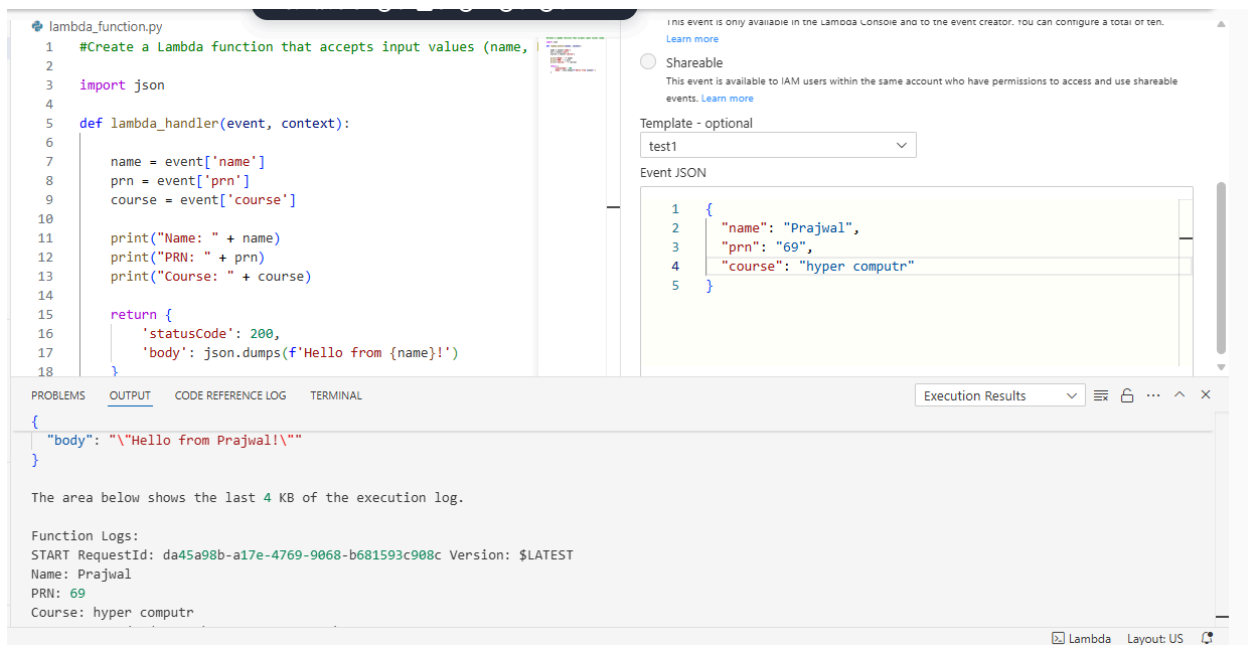


Assignment 8

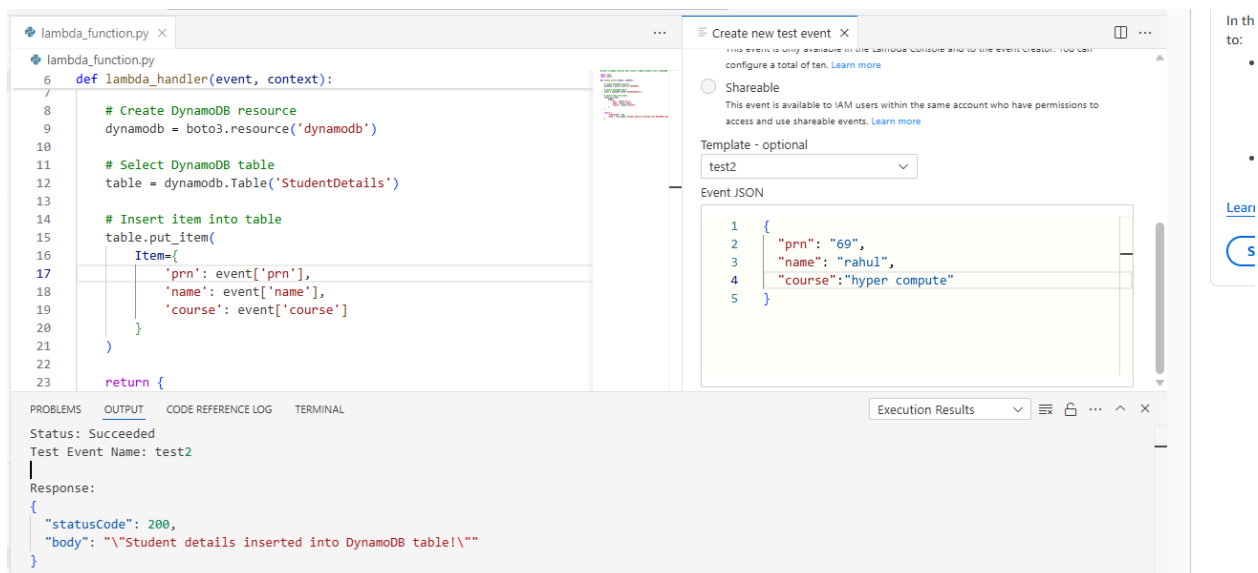
1. Create a AWS Lambda function that takes two numbers as input using a test event and returns their addition.



2. Create a Lambda function that accepts input values (name, PRN, course) from a test event and prints them.



3. Create a Lambda function that inserts student details into a DynamoDB table.



Completed · Items returned: 1 · Items scanned: 1 · Efficiency: 100% · RCUs consumed: 2

Table: StudentDetails - Items returned (1)

Scan started on April 29, 2026, 10:46:18

☐

prn (String)

▼

☐

69

course

▼

name

▼

hyper compute

rahul

Actions

Create item

<

1

>

⚙️

4. Create a Lambda function that reads student details from DynamoDB using PRN.

lambda_function.py

```

6  def lambda_handler(event, context):
7
8      prn = event['prn']
9
10     # Create DynamoDB resource
11     dynamodb = boto3.resource('dynamodb')
12
13     # Select the table
14     table = dynamodb.Table('StudentDetails')
15
16     # Get the item from DynamoDB
17     response = table.get_item(
18         Key={
19             'prn': prn
20         }
21     )
22
23     # Return the item
24     return {
25         'statusCode': 200,
26         'body': json.dumps(response['Item'])
27     }

```

PROBLEMS

OUTPUT

CODE REFERENCE LOG

TERMINAL

Status: Succeeded

Test Event Name: read1

Response:

```

{
  "statusCode": 200,
  "body": "{\"name\": \"rahul\", \"prn\": \"69\", \"course\": \"hyper compute\"}"
}

```

5. Create a Lambda function that updates a student's course name in DynamoDB.

before Updating:

☐	70	space computing	mausam
☐	69	hyper compute	rahul

code:

```

lambda_function.py
6 def lambda_handler(event, context):
12     table = dynamodb.Table('StudentDetails')
13
14     # Get student ID and new course name from the event
15     prn = event['prn']
16     new_course = event['new_course']
17
18     # Update the course name for the specified student ID
19     response = table.update_item(
20         Key={
21             'prn': prn
22         },
23         UpdateExpression='SET course = :val1',
24         ExpressionAttributeValues={
25             ':val1': new_course
26         }
27     )
28
29     return {
30         'statusCode': 200,
31         'body': json.dumps('Course updated successfully!')}

```

test2

Event JSON

```

1 {
2   "prn": "70",
3   "new_course": "ultra alien computing"
4 }
5

```

PROBLEMS OUTPUT CODE REFERENCE LOG TERMINAL

Status: Succeeded

Test Event Name: test2

Response:

```

{
  "statusCode": 200,
  "body": "\"Course updated successfully!\""
}

```

after Updating:

☐	70	ultra alien computing	mausam
☐	69	hyper compute	rahul

6. Create a Lambda function that deletes a student record from DynamoDB using PRN.

Table:

Table: StudentDetails - Items returned (2)			
Scan started on April 29, 2026, 11:47:27			
☐	prn (String)	course	name
☐	70	ultra alien computing	mausam
☐	69	hyper compute	rahul

Code:

lambda_function.py

```

3 import json
4 import boto3
5
6
7 def lambda_handler(event, context):
8
9     prn=event['prn']
10
11     dynamodb = boto3.resource('dynamodb')
12     table = dynamodb.Table('StudentDetails')
13
14     response = table.delete_item(
15         Key={
16             'prn': prn
17         }
18     )
19
20     return {
21         'statusCode': 200,
22         'body': json.dumps('Student record deleted successfully!')
23     }

```

Create new test event

Event sharing settings

Private

This event is only available in the Lambda Console and to the event creator. You can configure a total of ten. [Learn more](#)

Shareable

This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional

test1

Event JSON

```

1 {
2   "prn": "70"
3 }

```

PROBLEMS

OUTPUT

CODE REFERENCE LOG

TERMINAL

Execution Results

Response:

```

{
  "statusCode": 200,
  "body": "\"Student record deleted successfully!\""
}

```

After Deletion:

Completed · Items returned: 1 · Items scanned: 1 · Efficiency: 100% · RCUs consumed: 2			
Table: StudentDetails - Items returned (1)			
Scan started on April 29, 2026, 11:57:10			
☐	prn (String)	course	name
☐	69	hyper comp...	rahul

7. Create a Lambda function that scans all student records from a DynamoDB table.

lambda_function.py

```

2
3 import json
4 import boto3
5
6 def lambda_handler(event, context):
7
8     # Create a DynamoDB resource
9     dynamodb = boto3.resource('dynamodb')
10
11     # Select the DynamoDB table
12     table = dynamodb.Table('StudentDetails')
13
14     # Scan the table
15     response = table.scan()
16
17     # Return the items
18     return {
19         'statusCode': 200,
20         'body': json.dumps(response['Items'])
21     }
22

```

Event sharing settings

☒ Private
This event is only available in the Lambda Console and to the event creator. You can configure a total of ten. [Learn more](#)

☐ Shareable
This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional
test1

Event JSON

```

1 {
2
3 }

```

PROBLEMS OUTPUT CODE REFERENCE LOG TERMINAL

Execution Results

Response:

```

{
  "statusCode": 200,
  "body": "[{\\"name\\": \\"rahul\\", \\"prn\\": \\"69\\", \\"course\\": \\"hyper compute\\"}]"
}

```

The area below shows the last 4 KB of the execution log.

8. Create a Lambda function that checks whether a student PRN already exists before inserting a new record.

Table: StudentDetails - Items returned (1)

Actions

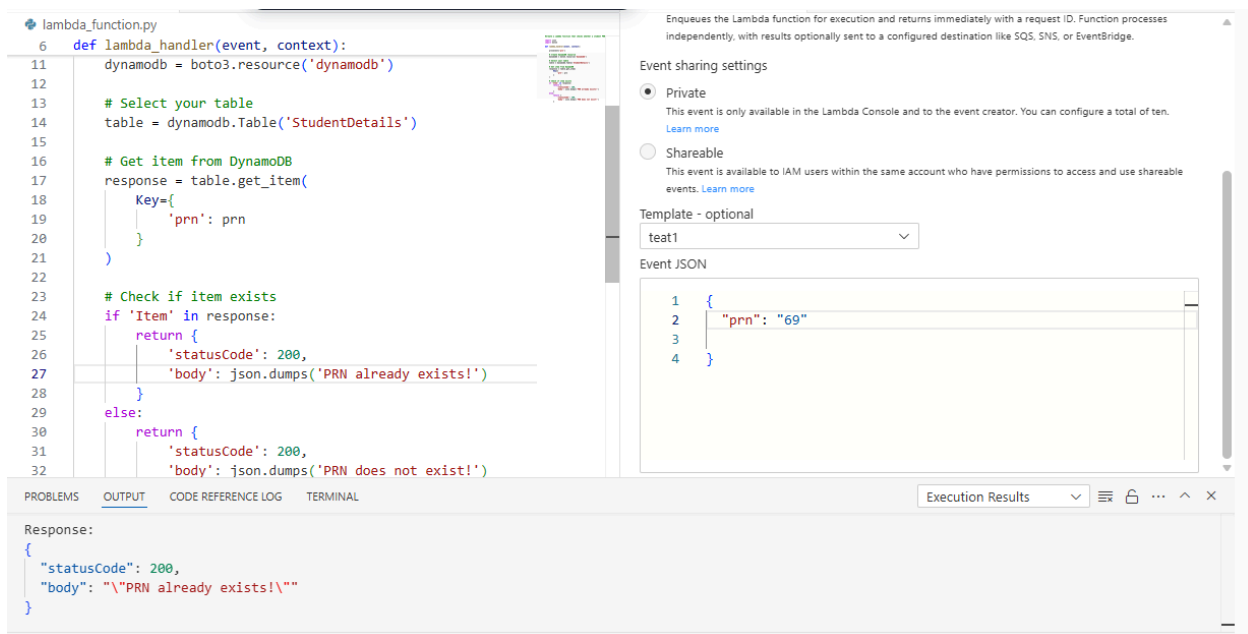
Create item

Scan started on April 29, 2026, 11:57:10

< 1 >

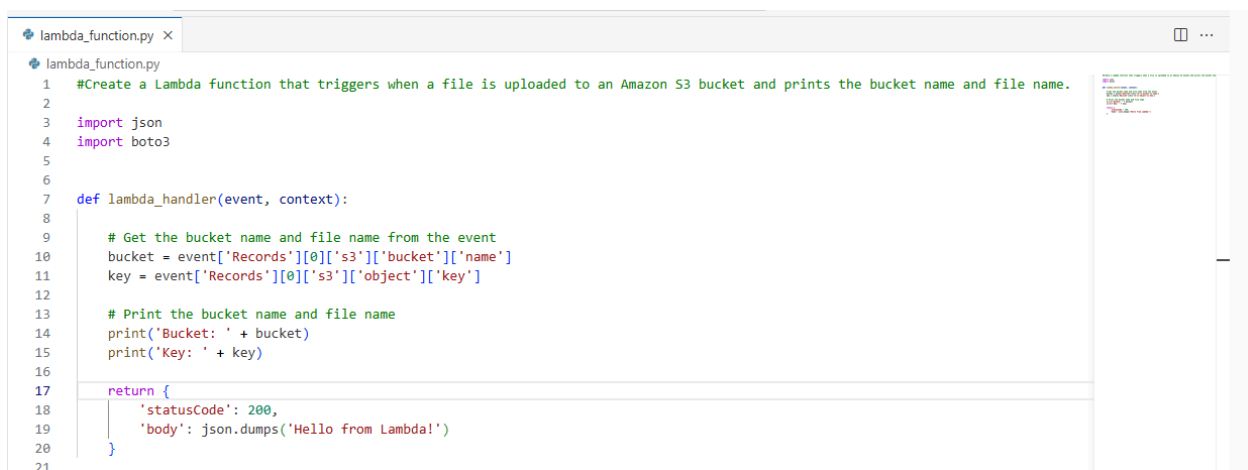
⚙

☐	prn (String)	course	name
☐	69	hyp...	rahul



9. Create a Lambda function that triggers when a file is uploaded to an Amazon S3 bucket and prints the bucket name and file name.

CODE:



Trigger:

Function overview Info

Diagram

Template

S3

+ Add trigger

MonitorS3

Layers (0)

+ Add destination

Export to Infrastructure Composer

Download

Function ARN

arn:aws:lambda:us-east-1:107970445632:function:MonitorS3

Description

-

Last modified

3 minutes ago

S3 file upload:

Upload succeeded

For more information, see the Files and folders table.

×

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://lamdatestbucket69

Succeeded

1 file, 11.7 KB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 11.7 KB)

Find by name

Name	Folder	Type	Size	Status	Error
game.html	-	text/html	11.7 KB	Succeeded	-

Logs:

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms

AI Operations

GenAI Observability

Application Signals (APM)

Infrastructure Monitoring

Logs

Log Management

Log management

/aws/lambda/MonitorS3

2026/04/29/[\$LATEST]5f10cc32b374d71852c805fc06d5c48

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp

Message

No older events at this moment. [Retry](#)

2026-04-29T07:13:48.636Z

INIT_START Runtime Version: python:3.14.v99 Runtime Version ARN: arn:aws:lambda:us-east-1:runtime:86dae9a05ccfc050de4f62a6c3f8c397642962ec872732763a914c004131809

2026-04-29T07:13:48.963Z

START RequestId: 621f124e-d571-40ab-81d9-130fd364c38e Version: \$LATEST

2026-04-29T07:13:48.962Z

Bucket: lamdatestbucket69

2026-04-29T07:13:48.962Z

Key: game.html

2026-04-29T07:13:48.965Z

END RequestId: 621f124e-d571-40ab-81d9-130fd364c38e

2026-04-29T07:13:48.965Z

REPORT RequestId: 621f124e-d571-40ab-81d9-130fd364c38e Duration: 3.54 ms Billed Duration: 329 ms Memory Size: 128 MB Max Memory Used: 69 MB Init Duration: 324.53 ms

No newer events at this moment. [Auto retry paused. Resume](#)

10. Create a Lambda function that triggers on S3 upload and prints the size of the uploaded file.

code:

```
lambda_function.py X
lambda_function.py
6 def lambda_handler(event, context):
7
8     # Get the bucket name and object key from the event
9     bucket = event['Records'][0]['s3']['bucket']['name']
10    key = event['Records'][0]['s3']['object']['key']
11
12    # Create an S3 client
13    s3 = boto3.client('s3')
14
15    # Get the object metadata
16    response = s3.head_object(Bucket=bucket, Key=key)
17
18    # Get the file size
19    file_size = response['ContentLength']
20
21    # Print the file size
22    print(f"File size: {file_size} bytes")
23
24    return {
25        'statusCode': 200,
26        'body': json.dumps(f'File size: {file_size} bytes')
27    }
28
```

trigger:

Trigger-Size-and-upload Throttle Copy ARN Actions

Function overview Info

Diagram Template

 **Trigger-Size-and-upload**

 Layers (0)

 S3

+ Add trigger

+ Add destination

Function ARN
arn:aws:lambda:us-east-1:107970445632:function:Trigger-Size-and-upload

Description
-

Last modified
6 seconds ago

Export to Infrastructure Composer Download

Uploaded on S3 bucket:

Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination s3://bucketsizer	Succeeded 1 file, 35.7 MB (100.00%)	Failed 0 files, 0 B (0%)
--	---	------------------------------------

Files and folders Configuration

Files and folders (1 total, 35.7 MB)

Find by name

Name	Folder	Type	Size	Status	Error
PacketTracer5.exe	-	application/x-msdownload	35.7 MB	Succeeded	-

Monitoring:

Log management > /aws/lambda/Trigger-Size-and-upload > 2026/04/29/[\$LATEST]7287990a6daa4217b035ff6d2c452892

< >

Log events

Actions Start tailing Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp	Message
No older events at this moment. Retry	
2026-04-29T07:24:03.504Z	INIT_START Runtime Version: python:3.14.v39 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:06dae9ab05ccfc05b6ef62a6c3f8c3976f2962ec872732763a014c0041151809
2026-04-29T07:24:03.836Z	START RequestId: e68b66a0-c174-4ce1-b183-50d490c679da Version: \$LATEST
2026-04-29T07:24:06.704Z	[ERROR] ClientError: An error occurred (403) when calling the HeadObject operation: Forbidden Traceback (most recent call last): File "/var/task/lambda_function.py", line 16, in lambda_han...
2026-04-29T07:24:06.725Z	END RequestId: e68b66a0-c174-4ce1-b183-50d490c679da
2026-04-29T07:24:06.725Z	REPORT RequestId: e68b66a0-c174-4ce1-b183-50d490c679da Duration: 2888.62 ms Billed Duration: 3221 ms Memory Size: 128 MB Max Memory Used: 96 MB Init Duration: 331.61 ms
2026-04-29T07:25:11.890Z	START RequestId: e68b66a0-c174-4ce1-b183-50d490c679da Version: \$LATEST
2026-04-29T07:25:12.183Z	[ERROR] ClientError: An error occurred (403) when calling the HeadObject operation: Forbidden Traceback (most recent call last): File "/var/task/lambda_function.py", line 16, in lambda_han...
2026-04-29T07:25:12.224Z	END RequestId: e68b66a0-c174-4ce1-b183-50d490c679da
2026-04-29T07:25:12.224Z	REPORT RequestId: e68b66a0-c174-4ce1-b183-50d490c679da Duration: 334.16 ms Billed Duration: 335 ms Memory Size: 128 MB Max Memory Used: 97 MB
2026-04-29T07:27:14.412Z	START RequestId: e68b66a0-c174-4ce1-b183-50d490c679da Version: \$LATEST
2026-04-29T07:27:14.679Z	[ERROR] ClientError: An error occurred (403) when calling the HeadObject operation: Forbidden Traceback (most recent call last): File "/var/task/lambda_function.py", line 16, in lambda_han...
2026-04-29T07:27:14.724Z	END RequestId: e68b66a0-c174-4ce1-b183-50d490c679da
2026-04-29T07:27:14.725Z	REPORT RequestId: e68b66a0-c174-4ce1-b183-50d490c679da Duration: 312.17 ms Billed Duration: 313 ms Memory Size: 128 MB Max Memory Used: 99 MB
No newer events at this moment. Auto retry paused. Resume	